

Project Scoping: Medical Device Verification Modules

1. Evidence Engine (Automated Validation Scoring)

Objective: To automate the retrieval, analysis, and scoring of peer-reviewed clinical evidence to substantiate medical device claims.

Key Functional Requirements:

- **Database Integration (API Handshakes):** Establish secure connections to major clinical registries and biomedical databases (e.g., PubMed, ClinicalTrials.gov, and WHO ICTRP).
- **Metadata Extraction:** Build parsers to identify critical study parameters, including sample size (n), study design (e.g., Randomized Controlled Trials vs. Retrospective studies), and demographic diversity.
- **Automated Scoring Logic:** Implement a weighted scoring engine based on established clinical hierarchies.
 - *High Weight:* Multi-site prospective trials, peer-reviewed validation.
 - *Medium Weight:* Single-site retrospective analyses.
 - *Low Weight:* Case studies or white papers.
- **Verification Ledger:** Create a structured output that links every device claim directly to its supporting citation and its associated "Evidence Score."

2. Performance Layer (Real-World Monitoring & Drift Detection)

Objective: To monitor the operational health of AI/ML devices in real-time and identify when a model's performance begins to deviate from its validated baseline.

Key Functional Requirements:

- **RWPM Ingestion Gateway:** A secure API layer designed to ingest Real-World Performance Monitoring (RWPM) telemetry from deployed medical devices.
- **Entropy & Drift Detection:** Implementation of statistical monitoring to identify **Algorithmic Drift**.
 - **Data Drift:** Identifying changes in the input data distribution that may invalidate the model's original training assumptions.
 - **Concept Drift:** Identifying shifts in the relationship between input data and the predicted outcome (e.g., changes in clinical diagnostic standards).
- **Safety Thresholding (PCCP Alignment):** Define specific "stability zones" based on a Predetermined Change Control Plan (PCCP).

- **Automated Alerting:** A trigger system that flags high-entropy states for Human-in-the-Loop (HITL) review before performance degradation impacts clinical outcomes.

Implementation Road Map

1. **Phase 1: API Discovery:** Identify and test authentication protocols for primary evidence databases.
2. **Phase 2: Schema Design:** Define the JSON structures for evidence scoring and performance telemetry.
3. **Phase 3: Prototype Build:** Develop a "Handshake Script" to fetch a single study and return an Evidence Score.
4. **Phase 4: Integration:** Bridge the two layers so that performance drift can be viewed alongside the original evidence base for a holistic safety view.